

Solution Code - Module 3-Core Java - ArraySumAndAverage

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q14_ArraySumAndAverage.java

```
1 | import java.util.Scanner;
2 |
3 | public class Q14_ArraySumAndAverage {
4 |     public static void main(String[] args) {
5 |         Scanner scanner = new Scanner(System.in);
6 |         System.out.print("Enter the number of elements: ");
7 |
8 |         if (scanner.hasNextInt()) {
9 |             int n = scanner.nextInt();
10 |             if (n <= 0) {
11 |                 System.out.println("Error: Number of elements must be greater than zero.");
12 |             } else {
13 |                 double[] array = new double[n];
14 |                 System.out.println("Enter " + n + " elements:");
15 |                 double sum = 0;
16 |                 boolean valid = true;
17 |
18 |                 for (int i = 0; i < n; i++) {
19 |                     System.out.print("Element " + (i + 1) + ": ");
20 |                     if (scanner.hasNextDouble()) {
21 |                         array[i] = scanner.nextDouble();
22 |                         sum += array[i];
23 |                     } else {
24 |                         System.out.println("Error: Invalid number.");
25 |                         valid = false;
26 |                         break;
27 |                     }
28 |                 }
29 |
30 |                 if (valid) {
31 |                     double average = sum / n;
32 |                     System.out.printf("Sum of elements: %.2f\n", sum);
33 |                     System.out.printf("Average of elements: %.2f\n", average);
34 |                 }
35 |             }
36 |         } else {
37 |             System.out.println("Error: Invalid number of elements.");
38 |         }
39 |
40 |         scanner.close();
41 |     }
42 | }
```